Chapter 3

Special Techniques

Problem 3.1

Find the average potential over a spherical surface of radius R due to a point charge q located inside (same as above, in other words, only with $z < R$). (In this case, of course, Laplace's equation does not hold within the sphere.) Show that, in general,

$$V_{\text{ave}} = V_{\text{center}} + \frac{Q_{\text{enc}}}{4\pi\epsilon_0 R}$$

where V_{center} is the potential at the center due to all the external charges, and Q_{enc} is the total enclosed charge.

Solution:

$$\boldsymbol{\imath} = R\hat{\mathbf{r}} - z\hat{\mathbf{z}}$$

$$\imath = \sqrt{R^2 + z^2 - 2Rz\cos\theta}$$

$$V_{\text{ave}} = \frac{1}{4\pi R^2}\frac{q}{4\pi\epsilon_0}\int \frac{1}{\sqrt{R^2 + z^2 - 2Rz\cos\theta}} R^2 \sin\theta \, \mathrm{d}\theta \, \mathrm{d}\varphi$$

$$\downarrow$$

$$V_{\text{ave}} = \frac{qR^2 2\pi}{4\pi R^2 4\pi\epsilon_0}\left[\frac{\sqrt{R^2 + z^2 - 2Rz\cos\theta}}{Rz}\right]_0^{\pi}$$

$$\downarrow$$

$$V_{\text{ave}} = \frac{q}{8\pi\epsilon_0 Rz}[(R+z) - (R-z)]$$

$$\downarrow$$

$$\boxed{V_{\text{ave}} = \frac{q}{4\pi\epsilon_0 R}}$$

Superposition gives that if there are more charges inside the sphere, V_{ave} is just the total charge (Q_{enc}) and therefore the general expression for V_{ave} is the contribution from chares outside (V_{center}) and the contribution from charges inside (V_{inside})

□

Problem 3.2

In one sentence, justify **Earnshaw's Theorem**: A charged particle cannot be held in a stable equilibrium by electrostatic forces alone. As an example, consider the cubical arrangement of fixed charges in Fig. 3.4. It looks, off hand, as though a positive charge at the center would be suspended in midair, since it is repelled away from each corner. Where is the leak in this "electrostatic bottle"? [To harness nuclear fusion as a practical energy source it is necessary to heat a plasma (soup of charged particles) to fantastic temperatures—so hot that contact would vaporize any ordinary pot. Earnshaw's theorem says that electrostatic containment is also out of the question. Fortunately, it is possible to confine a hot plasma magnetically.]

Solution:

There is no local maximum or minimum point in the solution to Laplace's equation. The apparent equilibrium point in the example must therefore be a saddle point. □

Problem 3.3

Find the general solution to Laplace's equation in spherical coordinates, for the case where V depends only on r. Do the same for cylindrical coordinates, assuming V depends only on s.

Solution:

$$\boldsymbol{\nabla}^2 V = 0$$

The laplacian in spherical coordinates with only r-dependence is

$$\boldsymbol{\nabla}^2 = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right)$$

$$\downarrow$$

$$\boldsymbol{\nabla}^2 V = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial V}{\partial r}\right) = \frac{1}{r^2}\left(2r\frac{\partial V}{\partial r} + r^2\frac{\partial^2 V}{\partial r^2}\right) = \frac{\partial^2 V}{\partial r^2} + \frac{2}{r}\frac{\partial V}{\partial r} = 0$$

This is Bessel diff.eq with $\lambda = 0, l = 0$

$$V = a + \frac{b}{r}$$

For the cylindrical part, the laplacian is

$$\boldsymbol{\nabla}^2 V = \frac{1}{s}\frac{\partial}{\partial s}\left(s\frac{\partial V}{\partial s}\right) = [\text{multiply by s and integrate}] = A$$

$$s\frac{\partial V}{\partial s} = A \rightarrow \frac{\partial V}{\partial s} = \frac{A}{s} \rightarrow V = A\ln(s) + B$$

□

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